

Data Mining :-

Data mining is the process of extracting useful patterns, relationship, trends and knowledge from large volumes of data using techniques such as machine learning.

- 1) Classification :- Assign data into predefined categories.
- 2) Clustering :- Group similar data objects together without predefined.

Steps involved in the KDD :-

- 1) Data Selection :- Collect relevant data from different data.
- 2) Data Cleaning :- Remove missing, incorrect, duplicate, or inconsistent data.
- 3) Data Integration :- Combine data from multiple databases into one.
- 4) Data mining :- Apply algorithms to discover useful patterns.
- 5) Pattern Evaluation :- Identify and evaluate meaningful and interesting patterns.

(b)

1) Missing values :-

Some data fields are empty or unavailable.
Ex:- Customer age is missing.

2) Noisy Data :-

Data contains errors, random variation, or outliers.

Ex:- Customer age entered as 250 years.

Example:- Age:- 22, 25, ?, 28

$$\text{Mean} = (22 + 25 + 28) / 3 = 25$$

Missing value $\rightarrow 25$

Data Integration
from multiple source into a

Challenges:-

1) Redundancy:-

The same data may exist in multiple datasets, causing duplicate records.

2) Inconsistency:-

Different sources may store the same information in different formats.

How Redundancy is Handled:-

1) Remove duplicate entries through data cleaning.

2) Use normalization to reduce repeated data.

Example:- Database A:- Customer ID = 101, Name = Chanan

Database B:- Customer ID = 101, Name = Chanan

Inconsistency is Handled:-

Database A: Date = 03/08/2026

Database B: Date = 2026-08-03

Both are converted into same format

(b)

Covariance Analysis:-

Covariance analysis is a statistical technique used to measure how two variables change together.

$$\text{Cov}(X, Y) = \frac{\sum_{i=1}^N (x_i - \mu_x)(y_i - \mu_y)}{N}$$

• N = Population size.

μ_x, μ_y = Population means.

Sample Covariance:-

$$= \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

n = Sample size

$\bar{x}, \bar{y}$ = Sample means.

$$A = (2, 3, 5, 4, 6)$$

$$B = (5, 8, 10, 11, 14)$$

$$\text{Mean of } A = 4$$

$$\text{Mean of } B = 9.6$$

$$\sum (x - \bar{x})(y - \bar{y}) = 20$$

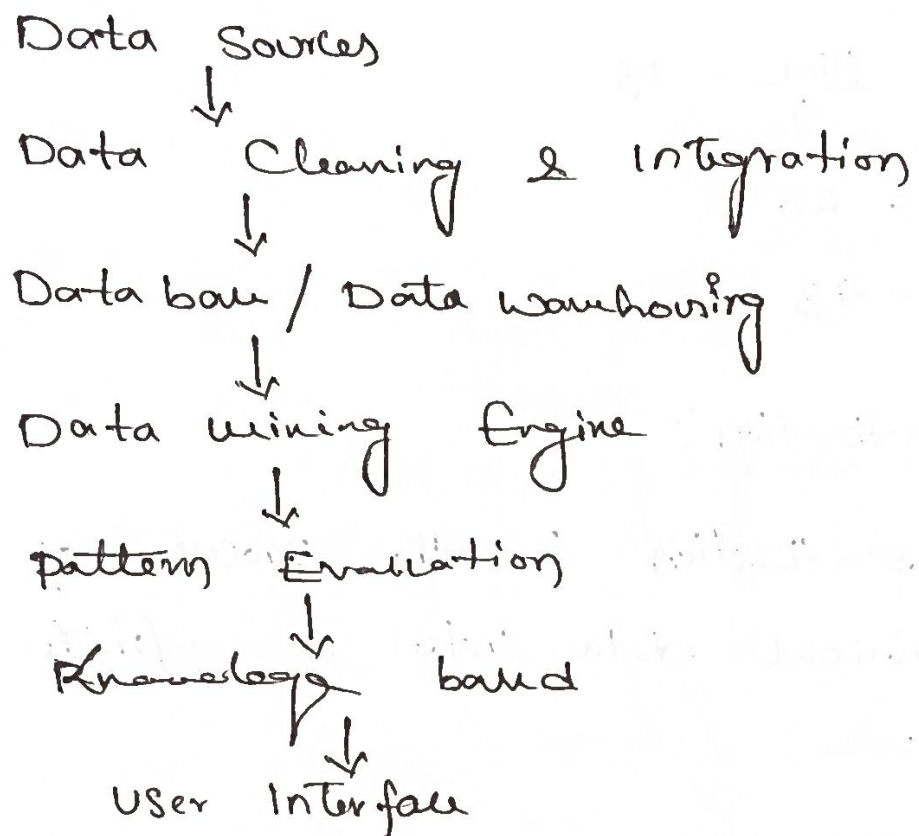
Sample Covariance

$$\text{Cov}(x, y) = \frac{20}{5-1} = 5$$

$$\text{Covariance} = 5 \text{ (positive)} //$$

(b)

Architecture Diagram:-



4) a) Given

$$x = \{4, 5, 6, 7, 9\}$$

$$y = \{8, 3, 7, 5, 6\}$$

$$\text{Mean of } X = \bar{x} = \frac{4+5+6+7+9}{5} = 6.2$$

$$\bar{y} = \frac{8+3+7+5+6}{5} = 5.8$$

$$\sum (x - \bar{x})(y - \bar{y}) = -1.8$$

$$= \frac{-1.8}{5.1} = -0.45 //$$

- 2) Data Integration
- 3) Data Transformation
- 4) Data Reduction
- 5) Data Discretization.

5.a) Given Data:

13, 15, 16, 16, 19, 20, 20, 21, 22, 25, 25, 25, 30, 33, 35, 35, 36,
40, 45, 46, 52, 70

Bin 1:- 13, 15, 16, 16, 19, 20, 20, 21, 22

Bin 2:- 22, 25, 25, 25, 25, 30, 33, 35

Bin 3:- 35, 35, 35, 36, 40, 45, 46, 52, 70

$$\text{mean} = \frac{\text{Bin 1}}{9} = 18$$

$$\text{Bin 2} = 28$$

$$\text{Bin 3} = 43$$

b) Data Discretization:-

Data Discretization is the process of converting continuous numerical data into a finite number of intervals (bins).

Importance:-

- Reduce data complexity.
- Removes noise.
- Improves mining and classification accuracy.
- Makes data easier to understand.